

Reading for Understanding

1. Why is the pharmaceutical industry showing interest in traditional folk remedies and in the neem tree in particular? (Paragraph 1) The pharmaceutical industry is interested in these remedies to aid the development of new drugs and pesticide products. The neem tree is of particular interest because people in the subcontinent have known for centuries that it is a "treasure trove" of medicinal and insecticidal chemicals.

2. What is the 'ancient practice' which dentists endorse? Why do they endorse it? (Paragraph 2) The "ancient practice" is using neem twigs to brush teeth. Dentists endorse this because it is highly effective in preventing gum inflammation and periodontal disease.

3. Explain in your own words why the neem tree is so important in the treatment of malaria. (Paragraph 2) Neem is vital because malaria remains a widespread and deadly disease, and the parasites that cause it are increasingly becoming resistant to standard synthetic drugs like chloroquine. Because traditional neem preparations have long been used as effective prophylactic drugs, researchers are returning to this natural source to find treatments that the parasites cannot yet resist.

4. Give three examples of the neem's 'chemical arsenal'. (Paragraph 3) The text describes the neem tree's chemical properties as:

- **Anti-viral:** Effective against viruses like chicken pox and measles.
- **Antiseptic:** Used to treat various skin infections and wounds.
- **Fungicidal:** Toxic to persistent fungi, such as those that cause athlete's foot.

5. What did the German entomologist observe in Sudan, and what could be learned from his observation? (Paragraph 4) During a massive locust plague, the entomologist observed that neem trees were the only green plants left completely untouched; the locust swarms flew off without feeding on them. This observation taught researchers that the neem tree protects itself with a natural "cocktail" of chemicals that act as powerful insect repellents, even against starving pests.

6. Why is neem wood and bark valued? (Paragraph 5)

- **Wood:** It is valued for being heavy, durable, and naturally resistant to termites and woodworm, making it ideal for furniture and construction.
- **Bark:** It provides a strong, coarse fiber used to make ropes and mats, and it also produces resins used in making glue.

7. Which parts of neem fruit are useful? What is produced from them? (Paragraph 6)

- **Seeds:** The kernels are crushed to produce "cold pressed" oil, which is used in soaps and cosmetics.
- **Residue (Neem cake):** The material left over after oil extraction is used as an excellent fertilizer or as feed for livestock and poultry.

